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ELECTROCHEMICAL CELL DEVICE, MODULE, AND MODULE HOUSING DEVICE

Abstract

The electrochemical cell device includes a cell, a support body, and a fixing member. The cell has a first end and extends from the first end in a first direction. The support body supports one end portion of the cell including the first end. The fixing member is located between the cell and the support body and is in contact with a first surface of the cell along the first direction and a second direction intersecting the first direction. The first surface includes a contact area in contact with the fixing member and a non-contact area not in contact with the fixing member. The contact area has a second end on the first end side. A length in the first direction from the second end to the non-contact area on a side opposite to the first end is larger at a second part located at an end portion in the second direction than at a first part located at a center portion in the second direction.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an electrochemical cell device, a module, and a module housing device.

BACKGROUND OF INVENTION

[0002] In recent years, various fuel cell stack devices each including a plurality of fuel cells have been proposed, as next-generation energy. A fuel cell is a type of electrochemical cell capable of obtaining electrical power by using a fuel gas such as a hydrogen-containing gas and an oxygen-containing gas such as air.

CITATION LIST

Patent Literature

[0003] Patent Document 1: JP 2015-35417 A

SUMMARY

[0004] An electrochemical cell device according to an aspect of the embodiment includes a cell, a support body, and a fixing member. The cell has a first end and extends from the first end in a first direction. The support body supports one end portion of the cell including the first end. The fixing member is located between the cell and the support body and is in contact with a first surface of the cell along the first direction and a second direction intersecting the first direction. The first surface includes a contact area in contact with the fixing member and a non-contact area not in contact with the fixing member. The contact area has a second end on a side of the first end. A length in the first direction from the second end to the non-contact area on a side opposite to the first end is larger at a second part located at an end portion in the second direction than at a first part located at a center portion in the second direction.

[0005] An electrochemical cell device of the present disclosure includes a first cell, a first support body, a second cell, a second support body, and a fixing member. The first cell has a first end and extends from the first end in a first direction. The first support body supports one end portion of the first cell including the first end. The second cell is adjacent to the first cell with the first support body interposed between the first cell and the second cell. The second support body supports one end portion of the second cell adjacent to the one end portion of the first cell. The fixing member is in contact with a first surface of the first cell along the first direction and a second direction intersecting the first direction, and a second surface of the second cell along the first direction and the second direction. The first surface includes a contact area in contact with the fixing member and a non-contact area not in contact with the fixing member. The contact area has a second end on a side of the first end. A length in the first direction from the second end to the non-contact area on a side opposite to the first end is larger at a second part located at an end portion in the second direction than at a first part located at a center portion in the second direction. A length, at the second part, in the first direction from the second end to the non-contact region on a side opposite to the first end is larger at a first end portion distant from the second cell than at a second end portion close to the second cell.

[0006] A module of the present disclosure includes the electrochemical cell device described above and a storage container housing the electrochemical cell device.

[0007] A module housing device of the present disclosure includes the module described above, an

auxiliary device configured to operate the module, and an external case housing the module and the auxiliary device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1A is a horizontal cross-sectional view illustrating an example of an electrochemical cell according to a first embodiment.

[0009] FIG. 1B is a side view of an example of the electrochemical cell according to the first embodiment when viewed from the side of an air electrode.

[0010] FIG. 1C is a side view of an example of the electrochemical cell according to the first embodiment when viewed from the side of an interconnector.

[0011] FIG. 2A is a perspective view illustrating an example of an electrochemical cell device according to the first embodiment.

[0012] FIG. 2B is a cross-sectional view taken along line X-X illustrated in FIG. 2A.

[0013] FIG. 2C is a top view illustrating an example of the electrochemical cell device according to the first embodiment.

[0014] FIG. 3 is a cross-sectional view taken along line Y-Y illustrated in FIG. 2C.

[0015] FIG. 4 is an enlarged cross-sectional view of a joined portion between a support body and an electrochemical cell in an electrochemical cell device according to a second embodiment.

[0016] FIG. 5 is an enlarged cross-sectional view of a joined portion between a support body and an electrochemical cell in an electrochemical cell device according to a third embodiment.

[0017] FIG. 6 is a cross-sectional view illustrating an electrochemical cell device according to a fourth embodiment.

[0018] FIG. 7 is a cross-sectional view illustrating an electrochemical cell device according to a fifth embodiment.

[0019] FIG. 8 is an exterior perspective view illustrating an example of a module according to an embodiment.

[0020] FIG. 9 is an exploded perspective view schematically illustrating an example of a module housing device according to an embodiment.

DESCRIPTION OF EMBODIMENTS

[0021] A fuel cell stack device includes, for example, a support body that supports a plurality of fuel cells. Such a structure has room for improvement in the durability of the joined portion between the support body and the fuel cell.

[0022] It is expected to provide an electrochemical cell device, a module, and a module housing device having high durability.

[0023] Embodiments of an electrochemical cell device, a module, and a module housing device disclosed in the present application will be described below in detail with reference to the accompanying drawings. Note that the disclosure is not limited by the following embodiments.

[0024] Note that the drawings are schematic and that the dimensional relationships between elements, the proportions of the elements, and the like may differ from the actual ones. There may be differences between the drawings in the dimensional relationships, proportions, and the like.

First Embodiment

Configuration of Electrochemical Cell

[0025] First, with reference to FIGS. 1A to 1C, an example of a solid oxide fuel cell will be described as an electrochemical cell according to the embodiment. The electrochemical cell device may include a cell stack including a plurality of the electrochemical cells. The electrochemical cell device including the plurality of electrochemical cells is simply referred to as a cell stack device.

[0026] FIG. 1A is a horizontal cross-sectional view illustrating an example of an electrochemical

cell according to the first embodiment, FIG. 1B is a side view of an example of the electrochemical cell according to the first embodiment when viewed from an air electrode side, and FIG. 1C is a side view of an example of the electrochemical cell according to the first embodiment when viewed from an interconnector side. Note that FIGS. 1A to 1C each illustrate an enlarged view of a part of a respective one of configurations of the electrochemical cell. Hereinafter, the electrochemical cell may be simply referred to as a cell.

[0027] In the example illustrated in FIGS. 1A to 1C, the cell 1 is hollow flat plate-shaped, and has an elongated plate shape. As illustrated in FIG. 1B, the shape of the entire cell 1 when viewed from the side is a rectangle having a side length of, for example, 5 cm to 50 cm in a length direction L and a length of, for example, 1 cm to 10 cm in a width direction W orthogonal to the length direction L. The thickness of the entire cell 1 in a thickness direction T is, for example, 1 mm to 5 mm. The length direction L is an example of a first direction. The width direction W is an example of a second direction intersecting the first direction.

[0028] As illustrated in FIG. 1A, the cell 1 includes a support substrate 2 with electrical conductivity, an element portion 3, and an interconnector 4. The support substrate 2 has a pillar shape having a pair of a first surface n1 and a second surface n2 that face each other, and a pair of arc-shaped side surfaces m that connect the first surface n1 and the second surface n2.

[0029] The element portion 3 is located on the first surface n1 of the support substrate 2. The element portion 3 includes a fuel electrode 5, a solid electrolyte layer 6, and an air electrode 8. In the example illustrated in FIG. 1A, the interconnector 4 is located on the second surface n2 of the cell 1. Note that the cell 1 may include an intermediate layer 7 between the solid electrolyte layer 6 and the air electrode 8.

[0030] As illustrated in FIG. 1B, the air electrode 8 does not extend to the lower end of the cell 1. At the lower end portion of the cell 1, only the solid electrolyte layer 6 is exposed on a surface of the first surface n1. As illustrated in FIG. 1C, the interconnector 4 may extend to the lower end of the cell 1. At a lower end portion of the cell 1, the interconnector 4 and the solid electrolyte layer 6 are exposed on the surface. Note that, as illustrated in FIG. 1A, the solid electrolyte layer 6 is exposed at the surface at the pair of side surfaces m in a circular arc shape of the cell 1. The interconnector 4 need not extend to the lower end of the cell 1.

[0031] Hereinafter, each of constituent members constituting the cell 1 will be described.

[0032] The support substrate 2 includes gas-flow passages 2a, in which gas flows. The example of the support substrate 2 illustrated in FIG. 1A includes six gas-flow passages 2a. The support substrate 2 has gas permeability, and allows the gas flowing in the gas-flow passage 2a to permeate to the fuel electrode 5. The support substrate 2 may have electrical conductivity. The support substrate 2 having electrical conductivity causes electricity generated in the element portion to be collected in the interconnector 4.

[0033] The material of the support substrate 2 includes, for example, an iron group metal component and an inorganic oxide. For example, the iron group metal component may be, for example, Ni (nickel) and/or NiO. The inorganic oxide may be, for example, a specific rare earth element oxide. The rare earth element oxide may contain, for example, one or more rare earth elements selected from Sc, Y, La, Nd, Sm, Gd, Dy, and Yb.

[0034] As the material of the fuel electrode 5, a commonly known material may be used. As the fuel electrode 5, porous electrically conductive ceramics, for example, ceramics containing: ZrO.sub.2 in which a calcium oxide, a magnesium oxide, or a rare earth element oxide is in solid solution, and Ni and/or NiO may be used. This rare earth element oxide may contain a plurality of rare earth elements selected from, for example, Sc, Y, La, Nd, Sm, Gd, Dy, and Yb. Hereinafter, ZrO.sub.2 in which a calcium oxide, a magnesium oxide, or a rare earth element oxide is contained as a solid solution may be referred to as stabilized zirconia. Stabilized zirconia may include partially stabilized zirconia.

[0035] The solid electrolyte layer 6 is an electrolyte and delivers ions between the fuel electrode 5

and the air electrode **8**. At the same time, the solid electrolyte layer **6** has gas blocking properties, and makes leakage of the fuel gas and the oxygen-containing gas less likely to occur.

[0036] The material of the solid electrolyte layer **6** may be, for example, ZrO_2 in which from 3 mole % to 15 mole % of a rare earth element oxide is in solid solution. The rare earth element oxide may contain, for example, one or more rare earth elements selected from Sc, Y, La, Nd, Sm, Gd, Dy, and Yb. The solid electrolyte layer **6** may contain, for example, ZrO_2 in which Yb, Sc, or Gd is in solid solution, CeO_2 in which La, Nd, or Yb is in solid solution, BaZrO_3 in which Sc or Yb is in solid solution, or BaCeO_3 in which Sc or Yb is in solid solution.

[0037] The air electrode **8** has gas permeability. The open porosity of the air electrode **8** may be, for example, in the range of 20% to 50%, particularly in the range of 30% to 50%. The open porosity of the air electrode **8** may also be referred to as the porosity of the air electrode **8**.

[0038] The material of the air electrode **8** is not particularly limited, as long as the material is one generally used for the air electrode. The material of the air electrode **8** may be, for example, an electrically conductive ceramic such as a so-called ABO₃ type perovskite oxide.

[0039] The material of the air electrode **8** may be, for example, a composite oxide in which strontium (Sr) and lanthanum (La) coexist at the A site. Examples of such a composite oxide include $\text{La}_x\text{Sr}_{1-x}\text{Co}_y\text{Fe}_{1-y}\text{O}_3$, $\text{La}_x\text{Sr}_{1-x}\text{MnO}_3$, $\text{La}_x\text{Sr}_{1-x}\text{FeO}_3$, and $\text{La}_x\text{Sr}_{1-x}\text{CoO}_3$. Here, x is $0 < x < 1$, and y is $0 < y < 1$.

[0040] When the element portion **3** includes the intermediate layer **7**, the intermediate layer **7** functions as a diffusion suppression layer. When strontium (Sr) contained in the air electrode **8** diffuses into the solid electrolyte layer **6**, a resistance layer of SrZrO_3 is formed in the solid electrolyte layer **6**. The intermediate layer **7** makes it difficult for Sr to diffuse, thereby making it difficult for SrZrO_3 to be formed.

[0041] A material of the intermediate layer **7** is not particularly limited as long as the material is one generally used for the diffusion suppression layer of an element between the air electrode **8** and the solid electrolyte layer **6**. The material of the intermediate layer **7** may contain, for example, CeO_2 (cerium oxide) in which rare earth elements other than Ce (cerium) are in solid solution. As such rare earth elements, for example, Gd (gadolinium), Sm (samarium), or the like may be used.

[0042] The interconnector **4** is dense, and makes the leakage of the fuel gas flowing through the gas-flow passages **2a** located inside the support substrate **2**, and of the oxygen-containing gas flowing outside the support substrate **2** less likely to occur. The interconnector **4** may have a relative density of 93% or more; particularly 95% or more.

[0043] As the material of the interconnector **4**, a lanthanum chromite-based perovskite oxide (LaCrO_3 -based oxide), a lanthanum strontium titanium-based perovskite oxide (LaSrTiO_3 -based oxide), or the like may be used. These materials have electrical conductivity, and are unlikely to be reduced and also unlikely to be oxidized even when brought into contact with a fuel gas such as a hydrogen-containing gas and an oxygen-containing gas such as air.

Configuration of Electrochemical Cell Device

[0044] An electrochemical cell device according to the present embodiment using the cell **1** described above will be described with reference to FIGS. **2A** to **2C**. FIG. **2A** is a perspective view illustrating an example of the electrochemical cell device according to the first embodiment, FIG. **2B** is a cross-sectional view taken along line X-X illustrated in FIG. **2A**, and FIG. **2C** is a top view illustrating an example of the electrochemical cell device according to the first embodiment.

[0045] As illustrated in FIG. **2A**, a cell stack device **10** includes a cell stack **11** including a plurality of the cells **1** arrayed (stacked) in the thickness direction T (see FIG. **1A**) of each cell **1**, and a fixing member **12**.

[0046] The fixing member **12** includes a fixing member **13** and a support member **14**. The support member **14** supports the cells **1**. The fixing member **13** fixes the cells **1** to the support member **14**. The support member **14** includes a support body **15** and a gas tank **16**. The support body **15** and the

gas tank **16**, constituting the support member **14**, are made of metal and electrically conductive.

[0047] As illustrated in FIG. 2B, the support body **15** includes an insertion hole **15a**, into which the lower end portions of the plurality of cells **1** are inserted. The lower end portions of the plurality of cells **1** and the inner wall of the insertion hole **15a** are bonded by the fixing member **13**.

[0048] The gas tank **16** includes an opening portion through which a reactive gas is supplied to the plurality of cells **1** via the insertion hole **15a**, and a recessed groove **16a** located in the periphery of the opening portion. The outer peripheral end portion of the support body **15** is bonded to the gas tank **16** by a bonding member **21**, with which the recessed groove **16a** of the gas tank **16** is filled.

[0049] In the example illustrated in FIG. 2A, the fuel gas is stored in an internal space **22** formed by the support body **15** and the gas tank **16**. The support body **15** and the gas tank **16** constitute the support member **14**. The gas tank **16** includes a gas circulation pipe **20** connected thereto. The fuel gas is supplied to the gas tank **16** through the gas circulation pipe **20** and is supplied from the gas tank **16** to the gas-flow passages **2a** (see FIG. 1A) inside the cells **1**. The fuel gas supplied to the gas tank **16** is generated by a reformer **102** (see FIG. 8) to be described later.

[0050] A hydrogen-rich fuel gas can be produced, for example, by steam-reforming a raw fuel. When the fuel gas is produced by steam reforming, the fuel gas contains steam.

[0051] The example illustrated in FIG. 2A includes two rows of cell stacks **11**, two support bodies **15**, and the gas tank **16**. The two rows of the cell stacks **11** each have a plurality of cells **1**. Each of the cell stacks **11** is fixed to a corresponding one of the support bodies **15**. An upper surface of the gas tank **16** includes two through holes. Each of the support bodies **15** is disposed in a corresponding one of the through holes. The internal space **22** is formed by the one gas tank **16** and the two support bodies **15**.

[0052] The insertion hole **15a** has, for example, an oval shape in a top surface view. The length of the insertion hole **15a** in an arrangement direction of the cells **1**, that is, the thickness direction T, may be longer than the distance between two end current collection members **17** located at both ends of the cell stack **11**, for example. The width of the insertion hole **15a** may be, for example, greater than the length of the cell **1** in the width direction W (see FIG. 1A).

[0053] As illustrated in FIG. 2B, the joined portions between the inner wall of the insertion hole **15a** and the lower end portions of the cells **1** are filled with the fixing member **13** and solidified. As a result, the inner wall of the insertion hole **15a** and the lower end portions of the plurality of cells **1** are bonded and fixed, and the lower end portions of the cells **1** are bonded and fixed to each other. The gas-flow passages **2a** of each of the cells **1** communicate, at the lower end portion, with the internal space **22** of the support member **14**.

[0054] The fixing member **13** and the bonding member **21** may be of low electrical conductivity, such as glass. As the specific materials of the fixing member **13** and the bonding member **21**, amorphous glass or the like may be used, and especially, crystallized glass or the like may be used.

[0055] As the crystallized glass, for example, any one selected from the group consisting of SiO.sub.2—CaO-based, MgO—B.sub.2O.sub.3-based, La.sub.2O.sub.3—B.sub.2O.sub.3—MgO-based, La.sub.2O.sub.3—B.sub.2O.sub.3—ZnO-based, and SiO.sub.2—CaO—ZnO-based materials may be used, or, in particular, a SiO.sub.2—MgO-based material may be used.

[0056] As illustrated in FIG. 2B, electrically conductive members **18** are each interposed between adjacent ones of the cells **1** of the plurality of cells **1**. Each of the electrically conductive members **18** electrically connects in series the fuel electrode **5** of one of the adjacent ones of the cells **1** with the air electrode **8** of the other one of the adjacent ones of the cells **1**. More specifically, the conductive member **18** electrically connects the interconnector **4** electrically connected to the fuel electrode **5** of the one of the adjacent cells **1** and the air electrode **8** of the other one of the adjacent cells **1**. The electrically conductive member **18** may be in contact with the fixing member **13** or may not be in contact with the fixing member **13**.

[0057] As illustrated in FIG. 2B, the end current collection members **17** are electrically connected to the cells **1** located at the outermost sides in the arrangement direction of the plurality of cells **1**.

The end current collection members **17** are each connected to an electrically conductive portion **19** protruding outward from the cell stack **11**. The electrically conductive portion **19** collects electricity generated by the cells **1**, and conducts the electricity to the outside. Note that in FIG. 2A, the end current collection members **17** are not illustrated.

[0058] As illustrated in FIG. 2C, in the cell stack device **10**, two cell stacks **11A** and **11B**, which are connected in series, function as one battery. Thus, the electrically conductive portion **19** of the cell stack device **10** is divided into a positive electrode terminal **19A**, a negative electrode terminal **19B**, and a connection terminal **19C**.

[0059] The positive electrode terminal **19A** functions as a positive electrode when the electrical power generated by the cell stack **11** is output to the outside, and is electrically connected to the end current collection member **17** on a positive electrode side in the cell stack **11A**. The negative electrode terminal **19B** functions as a negative electrode when the electrical power generated by the cell stack **11** is output to the outside, and is electrically connected to the end current collection member **17** on a negative electrode side in the cell stack **11B**.

[0060] The connection terminal **19C** electrically connects the end current collection member **17** on the negative electrode side in the cell stack **11A** and the end current collection member **17** on the positive electrode side in the cell stack **11B**.

Joining of Support Body and Electrochemical Cell

[0061] Next, joining between the support body **15** and the electrochemical cell will be described with reference to FIGS. 2B to 3. FIG. 3 is a cross-sectional view taken along line Y-Y illustrated in FIG. 2C.

[0062] As illustrated in FIG. 3, the support body **15** and the cell **1** are bonded to each other via the fixing member **13**. The fixing member **13** is located on a first end **1e** side which is a lower end of the cell **1** in the length direction L, and the support body **15** supports one end portion of the cell **1** including the first end **1e**. In FIG. 3, the shape and structure of the cell **1** are simplified.

[0063] As illustrated in FIG. 2B, the fixing member **13** is located between the cells **1** adjacent to each other in the thickness direction T (see FIG. 1A). As illustrated in FIG. 3, the fixing member **13** is located to be in contact with the first surface **1b** of the cell **1** along the length direction L and the width direction W. The first surface **1b** includes a contact area **31** in contact with the fixing member **13** and a non-contact area **32** not in contact with the fixing member **13**. The contact area **31** has a second end **31e** on the first end **1e** side. In FIG. 3, the second end **31e** coincides with the first end **1e**, but they may not coincide with each other.

[0064] The cell **1** constituting the cell stack device **10** may receive an external force in the width direction W. In such a case, the fixing member **13** that fixes the cell **1** may crack due to stress concentration caused by the external force applied to the cell **1**, and the durability of the cell stack device **10** may be reduced.

[0065] In the embodiment, the thickness of the fixing member **13** in contact with the cell **1** is changed according to the ease of concentration of the stress, and the stress generated in the fixing member **13** is dispersed. In the embodiment, the length in the length direction L from the second end **31e** of the contact area **31** to the non-contact area **32** is larger at the second part **31b** located at the end portion in the width direction W than at the first part **31a** located at the center portion in the width direction W.

[0066] When the cell **1** receives an external force in the width direction W, stress is more likely to be concentrated on the fixing member **13** located at the end portion in the width direction W than on the fixing member **13** located at the center portion in the width direction W. The thickness of the fixing member **13** at the second part **31b** located at the end portion in the width direction W is made larger than the thickness of the fixing member **13** at the first part **31a** located at the center portion in the width direction W, whereby the stress applied to the fixing member **13** can be dispersed. Therefore, since the durability of the fixing member **13** can be enhanced, the durability of the electrochemical cell device can be enhanced. The thickness of the fixing member **13** at the first part

31a located at the center portion in the width direction W is the length L1 from the second end **31e** of the contact area **31** to the non-contact area **32**. The width of the fixing member **13** at the second part **31b** located at the end portion in the width direction W is the length L2 from the second end **31e** of the contact area **31** to the non-contact area **32**.

[0067] The length L2 of the contact area **31** located at the end portion in the width direction W may be, for example, about $1.1 \times L1 \leq L2 \leq 2.5 \times L1$, particularly about $1.4 \times L1 \leq L2 \leq 1.7 \times L1$, with reference to the length L1 of the contact area **31** located at the center portion in the width direction W. By defining the lengths L1 and L2 in this manner, even when the cell **1** receives an external force in the width direction W, the stress received by the fixing member **13** can be dispersed, and cracks are less likely to occur in the fixing member **13**.

Second to Fifth Embodiments

[0068] Electrochemical cell devices according to second to fifth embodiments will be described with reference to FIGS. 4 to 7. FIG. 4 is an enlarged cross-sectional view of a joined portion between a support body and an electrochemical cell in the electrochemical cell device according to the second embodiment. In FIG. 4, the second end **31e** coincides with the first end **1e**, but they may not coincide with each other.

[0069] As illustrated in FIG. 4, in the first surface **1b** of the cell **1**, the second part **31b** located at the end portion in the width direction W may be located closer to the first part **31a** than the end surfaces **1c** and **1d** in the width direction W of the cell **1**. As described above, the second part **31b** may be located distant from both ends of the first surface **1b** in the width direction W.

[0070] The first part **31a** located at the center portion in the width direction W may or may not be located at positions equally spaced from both ends of the first surface **1b** in the width direction W. The contact area **31** of the first surface **1b** may have a portion which is not in contact with the fixing member **13**.

[0071] The fixing member **13** located outside the cell **1** in the width direction W may be flat along the width direction W, or the surface of the fixing member **13** may be inclined with respect to the width direction W so that the thickness of the fixing member **13** varies stepwise.

[0072] The second sites **31b** located at both end portions in the width direction W may have the same or different lengths from the second end **31e** of the contact area **31** to the non-contact area **32**.

[0073] The electrochemical cell device according to the first and second embodiments may include, for example, only one cell **1** or two or more cells **1** satisfying the above-described conditions. All the cells **1** included in the electrochemical cell devices according to the first and second embodiments may satisfy the above-described conditions. Such an electrochemical cell device can increase the durability as compared with an electrochemical cell device that does not include the cell **1** satisfying the above-described conditions.

[0074] FIG. 5 is an enlarged cross-sectional view of a joined portion between a support body and an electrochemical cell in the electrochemical cell device according to the third embodiment. As illustrated in FIG. 5, in the case where a first cell **1A** and a second cell **1B** are arranged in the width direction W of the cell **1**, the length in the length direction L from a second end **31Ae** of the contact area **31** to the non-contact area **32** in the second part **31b** of the first cell **1A** may be larger at the first end portion **31b1** distant from the second cell **1B** than at the second end portion **31b2** close to the second cell **1B**. The length from the second end **31Ae** of the contact area **31** to the non-contact area **32** at the first end portion **31b1** distant from the second cell **1B** is referred to as a length L3, and the length from the second end **31Ae** of the contact area **31** to the non-contact area **32** at the second end portion **31b2** close to the second cell **1B** is referred to as a length L4.

[0075] For example, in the cell stack device **10** having the first cell **1A** and the second cell **1B** arranged in the width direction W, the first cell **1A** is more likely to receive an external force in the width direction W at a portion distant from the second cell **1B** than at a portion close to the second cell **1B**. For this reason, the stress that the fixing member **13** receives can be dispersed by making the length L3 at the first end portion **31b1** of the first surface **1Ab** close to the portion that is likely

to receive the external force in the width direction W larger than the length L4 at the second end portion **31b2** close to the portion that is unlikely to receive the external force in the width direction W in the first cell **1A**. As a result, cracks are less likely to occur in the fixing member **13** that fixes the first cell **1A**, and the durability of the fixing member **13** can be increased. Therefore, the durability of the cell stack device **10** can be increased.

[0076] Similarly, in the fixing member **13** that fixes the second cell **1B**, the length in the length direction L from the second end **31Be** of the contact area **31** to the non-contact area **32** in the second part **31b** of the second surface **1Bb** corresponding to the first surface **1Ab** may be larger at the second end portion close to the first cell **1A** than at the first end portion distant from the first cell **1A**. Accordingly, the stress received by the fixing member **13** that fixes the second cell **1B** can be dispersed, and cracks are unlikely to be generated in the fixing member **13**. Therefore, since the durability of the fixing member **13** that fixes the second cell **1B** can be enhanced, the durability of the cell stack device **10** can be enhanced. In FIG. 5, the second end **31Ae** coincides with the first end **1Ae**, and the second end **31Be** coincides with the first end **1Be**, but they may not coincide with each other.

[0077] The electrochemical cell device according to the third embodiment may include, for example, only one first cell **1A** or second cell **1B** satisfying the above-described conditions, or may include two or more first cells **1A** or second cells **1B** satisfying the above-described conditions. All of the first cells **1A** and the second cells **1B** included in the electrochemical cell device according to the third embodiment may satisfy the above-described condition. Such an electrochemical cell device can have higher durability than an electrochemical cell device that does not include the first cell **1A** or the second cell **1B** satisfying the above-described conditions.

[0078] FIG. 6 is a cross-sectional view illustrating the electrochemical cell device according to the fourth embodiment. FIG. 7 is a cross-sectional view illustrating the electrochemical cell device according to the fifth embodiment.

[0079] The cell stack device **10** illustrated in FIGS. 6 and 7 is different from the cell stack device **10** illustrated in FIG. 2B in that the lengths in the length direction L of the fixing members **13** that fix the plurality of cells **1** arranged in the thickness direction T are different in accordance with the distances from the end current collection members **17**.

[0080] As illustrated in FIG. 6, the fixing member **13** may be located such that the length along the length direction L decreases with increasing distance from the end current collection member **17**. The fixing member **13** may be located such that the length along the length direction L is larger at a position close to the end current collection member **17** than at a position far from the end current collection member **17**, that is, the center portion in the arrangement direction of the cell stack. Although the cell **1** close to the end current collection member **17** is likely to receive an external force, by varying the length of the fixing member **13** along the length direction L in this manner, for example, even when the cell **1** receives an external force in the thickness direction T, the stress received by the fixing member **13** located in the thickness direction T of the cell **1** can be dispersed. Therefore, since the durability of the fixing member **13** can be enhanced, the durability of the electrochemical cell device can be enhanced. The fixing member **13** located in the thickness direction T of the cell **1** is, in other words, the fixing member **13** located between the cells **1** arranged in the thickness direction T and located outside the cell stack in the thickness direction T.

[0081] As illustrated in FIG. 7, the fixing member **13** may be located such that the length along the length direction L increases with increasing distance from the end current collection member **17**. The fixing member **13** may be located such that the length along the length direction L is larger at a position far from the end current collection member **17**, that is, at the center portion in the arrangement direction of the cell stack than at a position close to the end current collection member **17**. By varying the length of the fixing member **13** along the length direction L in this manner, for example, even when the support body **15** and/or the gas tank **16** is twisted due to random vibration caused by the driving of the cell stack device **10**, the strain of the support body **15** and/or the gas

tank **16** can be reduced at the center portion in the arrangement direction of the cell stack, which is likely to be influenced by the twist. Therefore, since the durability of the fixing member **13** can be enhanced, the durability of the electrochemical cell device can be enhanced.

Module

[0082] A module **100** according to an embodiment of the present disclosure using the cell stack device **10** according to each of the above-described embodiments will be described with reference to FIG. **8**. FIG. **8** is an external perspective view illustrating the module according to the embodiment. FIG. **8** illustrates a state in which the front and rear surfaces that are part of a storage container **101** are removed, and the cell stack device **10** of a fuel cell housed in the container is taken out rearward.

[0083] As illustrated in FIG. **8**, the module **100** includes the storage container **101**, and the cell stack device **10** housed in the storage container **101**. The reformer **102** is disposed above the cell stack device **10**.

[0084] The reformer **102** generates a fuel gas by reforming a raw fuel such as natural gas and kerosene, and supplies the fuel gas to the cell **1**. The raw fuel is supplied to the reformer **102** through a raw fuel supply pipe **103**. Note that the reformer **102** may include a vaporizing unit **102a** for vaporizing water and a reforming unit **102b**. The reforming unit **102b** includes a reforming catalyst (not illustrated) for reforming the raw fuel into a fuel gas. Such a reformer **102** can perform steam reforming, which is a highly efficient reformation reaction.

[0085] The fuel gas generated by the reformer **102** is supplied to the gas-flow passages **2a** (see FIG. **1A**) of the cell **1** through the gas circulation pipe **20**, the gas tank **16**, and the support member **14**.

[0086] In the module **100** having the configuration mentioned above, the temperature in the module **100** during normal power generation is about from 500° C. to 1000° C. due to combustion of gas and power generation by the cell **1**.

[0087] As described above, the module **100** is configured by housing the cell stack device **10** having high durability, so that the module **100** having high durability can be obtained.

[0088] Module Housing Device FIG. **9** is an exploded perspective view illustrating an example of a module housing device according to an embodiment. A module housing device **110** according to the present embodiment includes an external case **111**, the module **100** illustrated in FIG. **8**, and an auxiliary device (not illustrated). The auxiliary device operates the module **100**. The module **100** and the auxiliary device are housed in the external case **111**. Note that in FIG. **9**, the configuration is partially omitted.

[0089] The external case **111** of the module housing device **110** illustrated in FIG. **9** includes a support **112** and an external plate **113**. A dividing plate **114** vertically partitions the interior of the external case **111**. The space above the dividing plate **114** in the external case **111** is a module housing chamber **115** for housing the module **100**. The space below the dividing plate **114** in the external case **111** is an auxiliary device housing chamber **116** for housing the auxiliary device configured to operate the module **100**. Note that in FIG. **9**, the auxiliary device housed in the auxiliary device housing chamber **116** is omitted.

[0090] The dividing plate **114** includes an air circulation hole **117** for causing air in the auxiliary device housing chamber **116** to flow into the module housing chamber **115** side. The external plate **113** constituting the module housing chamber **115** includes an exhaust hole **118** for discharging air inside the module housing chamber **115**.

[0091] In the module housing device **110**, the module housing device **110** having high durability can be obtained by providing the module **100** having high durability in the module housing chamber **115** as described above.

[0092] Note that, in the embodiment described above, the case where the support substrate of the hollow flat plate-shaped is used has been exemplified; however, the embodiment can also be applied to an electrochemical cell device using a cylindrical support substrate.

Other Embodiments

[0093] An electrochemical cell device according to other embodiments will be described.

[0094] In each of the above-described embodiments, a so-called “vertically striped type” electrochemical cell device, in which only one element portion including the fuel electrode layer, the solid electrolyte layer, and the air electrode layer is provided on the surface of the support substrate, is exemplified. However, the present disclosure can be applied to a horizontally striped type electrochemical cell device with an array of so-called “horizontally striped type” electrochemical cells, in which a plurality of element portions are provided on the surface of the support substrate at mutually separated locations, and adjacent element portions are electrically connected to each other.

[0095] In each of the embodiments described above, a fuel cell, a fuel cell stack device, a fuel cell module, and a fuel cell device have been illustrated as examples of the “electrochemical cell”, the “electrochemical cell device”, the “module”, and the “module housing device”; however, they may be an electrolytic cell, an electrolytic cell stack device, an electrolytic module, and an electrolytic device, respectively, as other examples. The electrolytic cell includes a first electrode layer and a second electrode layer, and decomposes water vapor into hydrogen and oxygen or decomposes carbon dioxide into carbon monoxide and oxygen by supplying electric power. Although an oxide ion conductor or a hydrogen ion conductor is shown as an example of the electrolyte material of the electrochemical cell in each of the above embodiments, the electrolyte material may be a hydroxide ion conductor. According to the electrolytic cell, electrolytic cell stack device, electrolytic module, and electrolytic device discussed above, durability can be improved.

[0096] While the present disclosure has been described in detail, the present disclosure is not limited to the aforementioned embodiment, and various changes, improvements, and the like can be made without departing from the gist of the present disclosure.

[0097] As described above, the electrochemical cell device according to the embodiment includes the cell **1**, the support body **15**, and the fixing member **13**. The cell **1** has a first end **1e** and extends in a first direction from the first end **1e**. The support body **15** supports one end portion of the cell including the first end **1e**. The fixing member **13** is located between the cell **1** and the support body **15**, and is in contact with the first surface **1b** of the cell **1** along the first direction and a second direction intersecting the first direction. The first surface **1b** includes a contact area **31** in contact with the fixing member **13** and a non-contact area **32** not in contact with the fixing member **13**. The contact area **31** has a second end **31e** on the first end **1e** side. The length in the first direction from the second end **31e** to the non-contact area **32** on the side opposite to the first end **1e** is larger at the second part **31b** located at the end portion in the second direction than at the first part **31a** located at the center portion in the second direction. Thus, since the durability of the fixing member **13** can be enhanced, the durability of the electrochemical cell device can be enhanced.

[0098] The electrochemical cell device according to the embodiment includes the first cell **1A**, the first support body, the second cell **1B**, the second support body, and the fixing member **13**. The first cell **1A** has a first end **1Ae** and extends in the first direction from the first end **1Ae**. The first support body supports one end portion of the first cell **1A** including the first end **1Ae**. The second cell **1B** is adjacent to the first cell **1A** with the first support body interposed therebetween. The second support body supports one end portion of the second cell **1B** adjacent to the one end portion of the first cell **1A**. The fixing member **13** is in contact with the first surface **1Ab** of the first cell **1A** along the first direction and a second direction intersecting the first direction and the second surface **1Bb** of the second cell **1B** along the first direction and the second direction. The first surface **1Ab** includes a contact area **31** in contact with the fixing member **13** and a non-contact area **32** not in contact with the fixing member **13**. The contact area **31** has a second end **31Ae** on the first end **1Ae** side. The length in the first direction from the second end **31Ae** to the non-contact area **32** on the side opposite to the first end **1Ae** is larger at the second part **31b** located at the end portion in the second direction than at the first part **31a** located at the center portion in the second

direction. The length, at the second part **31b**, in the first direction from the second end **31Ae** to the non-contact area **32** on the side opposite to the first end **1Ae** is larger at the first end portion **31b1** distant from the second cell **1B** than at the second end portion **31b2** close to the second cell **1B**. Thus, since the durability of the fixing member **13** can be enhanced, the durability of the electrochemical cell device can be enhanced.

[0099] The module **100** according to the embodiment includes the electrochemical cell device described above, and a storage container **101** that houses the electrochemical cell device. Thus, the module **100** having high durability can be obtained.

[0100] The module housing device **110** according to the embodiment includes the module **100** described above, the auxiliary device configured to operate the module **100**, and the external case housing the module **100** and the auxiliary device. Thus, the module housing device **110** having a high durability can be obtained.

[0101] Note that the embodiments disclosed herein are exemplary in all respects and not restrictive. The aforementioned embodiments can be embodied in a variety of forms. The aforementioned embodiments may be omitted, replaced, or changed in various forms without departing from the scope of the appended claims and the purpose thereof.

REFERENCE SIGNS

[0102] **1** Cell [0103] **3** Element portion [0104] **5** Fuel electrode [0105] **6** Solid electrolyte layer [0106] **7** Intermediate layer [0107] **8** Air electrode [0108] **10** Cell stack device [0109] **11** Cell stack [0110] **12** Fixing member [0111] **13** Fixing member [0112] **14** Support member [0113] **15** Support body [0114] **16** Gas tank [0115] **17** End current collection member [0116] **18** Electrically conductive member [0117] **100** Module [0118] **110** Module housing device

Claims

1. An electrochemical cell device comprising: a cell comprising: a first end, the cell extending from the first end in a first direction; a first surface along the first direction and a second direction intersecting the first direction; a support body supporting one end portion of the cell comprising the first end; and a fixing member located between the cell and the support body, the fixing member in contact with the first surface, wherein the first surface comprises: a contact area in contact with the fixing member; and a non-contact area in no contact with the fixing member, the contact area comprises a second end located on a side of the first end, and a length in the first direction from the second end to the non-contact area on a side opposite to the first end is larger at a second part located at an end portion of the first surface in the second direction than at a first part located at a center portion of the first surface in the second direction.

2. An electrochemical cell device comprising: a first cell comprising: a first end, the first cell extending from the first end in a first direction; a first surface along the first direction and a second direction intersecting the first direction; a first support body supporting one end portion of the first cell comprising the first end; a second cell adjacent to the first cell, the second cell comprising a second surface along the first direction and the second direction; a second support body supporting one end portion of the second cell adjacent to the one end portion of the first cell; and a fixing member in contact with the first surface, wherein the first surface comprises: a contact area in contact with the fixing member; and a non-contact area in no contact with the fixing member, the contact area comprises a second end located on a side of the first end, a length in the first direction from the second end to the non-contact area on a side opposite to the first end is larger at a second part that is located at an end portion of the first surface in the second direction than at a first part that is located at a center portion of the first surface in the second direction, and a length, in the first direction from the second end to the non-contact area on a side opposite to the first end is larger at a first end portion of the first surface distant from the second cell than at a second end portion of the first surface close to the second cell.

3. A module comprising: the electrochemical cell device according to claim **1**; and a storage container housing the electrochemical cell device.

4. A module housing device comprising: the module according to claim **3**; an auxiliary device configured to operate the module; and an external case housing the module and the auxiliary device.

5. A module comprising: the electrochemical cell device according to claim **2**; and a storage container housing the electrochemical cell device.

6. A module housing device comprising: the module according to claim **5**; an auxiliary device configured to operate the module; and an external case housing the module and the auxiliary device.
